D A N I E L A K I N W A L E

M E C H A N I C A L & A E R O S P A C E E N G I N E E R

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E D U C A T I O N

Cornell University, College of Engineering, Ithaca, NY

Master of Engineering, Mechanical & Aerospace Engineering | GPA: 3.79 **Jan 2026 – Dec 2026 (Expected)**

- Graduate Coursework: Dynamics of Flight Vehicles, Additive Manufacturing, Model-Based Systems Engineering

Bachelor of Science, Mechanical Engineering **Aug 2022 – May 2026**

- Coursework: Propulsion (Aircraft & Rockets), Fluid Mechanics, Heat Transfer, System Dynamics, Mechatronics, GD&T, DFMA

W O R K E X P E R I E N C E

Graduate Researcher, Aerospace ADVERSARY Lab, Cornell University, Ithaca, NY **Mar 2026 – Present**

- Design CubeSat structures in CAD, including 3D-printed primary structures, for a satellite testbed that evaluates space cybersecurity applications under the lab's Principal Investigator (PI)
- Develop servo-actuated attitude adjustment that points CubeSats at one another for satellite-to-satellite communication

Fluids Manufacturing Intern, Vast Space, Long Beach, CA **Jun 2026 – Sep 2026**

- Supported orbital tube welding (OTW) for flight-hardware tube production using Manufacturo (MES), Siemens NX, and Teamcenter
- Served as nonconformance (NC) point of contact for Fluids tube production, supporting NC disposition of flight hardware
- Redesigned the 3D-printed OTW support fixtures, which the team adopted as its standard welding process
- Stood up a new Fluids-dedicated tube-proofing setup, enabling proof testing to higher cleanliness requirements

Photolithography Equipment Engineering Intern, GlobalFoundries, Malta, NY **Jun 2025 – Aug 2025**

- Accelerated in-line wafer defect detection by retrofitting TEL and SEMES track tools in a 300mm fab with camera/DVR monitoring
- Cut part cost and enabled one-person service by redesigning a tool-panel fastener as a 3D-printed magnetic assembly in SolidWorks

Mechanical Engineering Intern, SRW Engineering & Architecture, Philadelphia, PA **May 2024 – Aug 2024**

- Produced AutoCAD floor plans for MEP infrastructure (outlets, piping, wiring, HVAC) across new, in-progress, and completed sites
- Maximized livable space within floor-height and envelope limits by subdividing large buildings into single-family units

Shock & Vibration Applications Engineering Intern, The VMC Group, Bloomingdale, NJ **Jun 2021 – Aug 2021**

- Designed and optimized 6DOF shock and random vibration isolation systems for aerospace and defense customers
- Verified margins of safety against mission shock and vibration profiles using nonlinear dynamic FEA and hand calculations
- Advised customers on isolator selection for mission-critical equipment in turbulent-flight, off-road, and blast environments

P R O J E C T S

Robotic Arm Integration (M.Eng Project), sponsored by ASML, Cornell University **Jan 2026 – Present**

- Redesigned the mounting approach and built support structures for an ASML reticle handling robot arm
- Integrating an optical safety light curtain with the ASML sponsor to stop the arm when personnel enter its range of motion
- Developing the arm's motion-control software and operator documentation for student use

L E A D E R S H I P E X P E R I E N C E

Crew Manager, Emerson Manufacturing Lab, Cornell University, Ithaca, NY **Jan 2024 – Present**

- Designed and deployed structured training and proficiency assessments across CNC (HAAS), mill, lathe, waterjet, and laser cutter
- Improved studio uptime, machine reliability, and safety by specifying replacement tooling and coordinating procurement

Composites Team Lead, Cornell Hyperloop (Student Design Team), Ithaca, NY **Feb 2023 – May 2026**

- Led end-to-end design and manufacture of a carbon fiber composite aeroshell, from CAD iteration through ANSYS FEA validation
- Directed a composites sub-team through SolidWorks, ANSYS, and CNC machining to deliver flight-weight primary structures
- Authored composite layup documentation and chemical safety protocols for the team

S K I L L S

CAD & Design: SolidWorks, Siemens NX, Fusion 360, AutoCAD, GD&T (ASME Y14.5), DFM/DFMA
Analysis: ANSYS Workbench FEA, nonlinear dynamic analysis, shock & random vibration, 6DOF modeling, margins of safety
Manufacturing: 3-axis CNC (HAAS), mill, lathe, waterjet, laser cutter, composite wet layup, FFF/SLA 3D printing, G-Code
Quality & PLM: NC disposition, orbital tube welding (OTW), cleanliness testing, Manufacturo (MES), Teamcenter, Jira, Confluence
Programming & Certifications: MATLAB, Python, Arduino, Java, JavaScript, SQL; OSHA 10-Hour, ASHRAE training

P R O F E S S I O N A L A S S O C I A T I O N S & E X T R A C U R R I C U L A R A C T I V I T I E S

Professional Associations: National Society of Black Engineers (NSBE)
Extracurricular Activities: Cornell Rock Climbing Club, Ski & Snowboard Club, Boxing Club